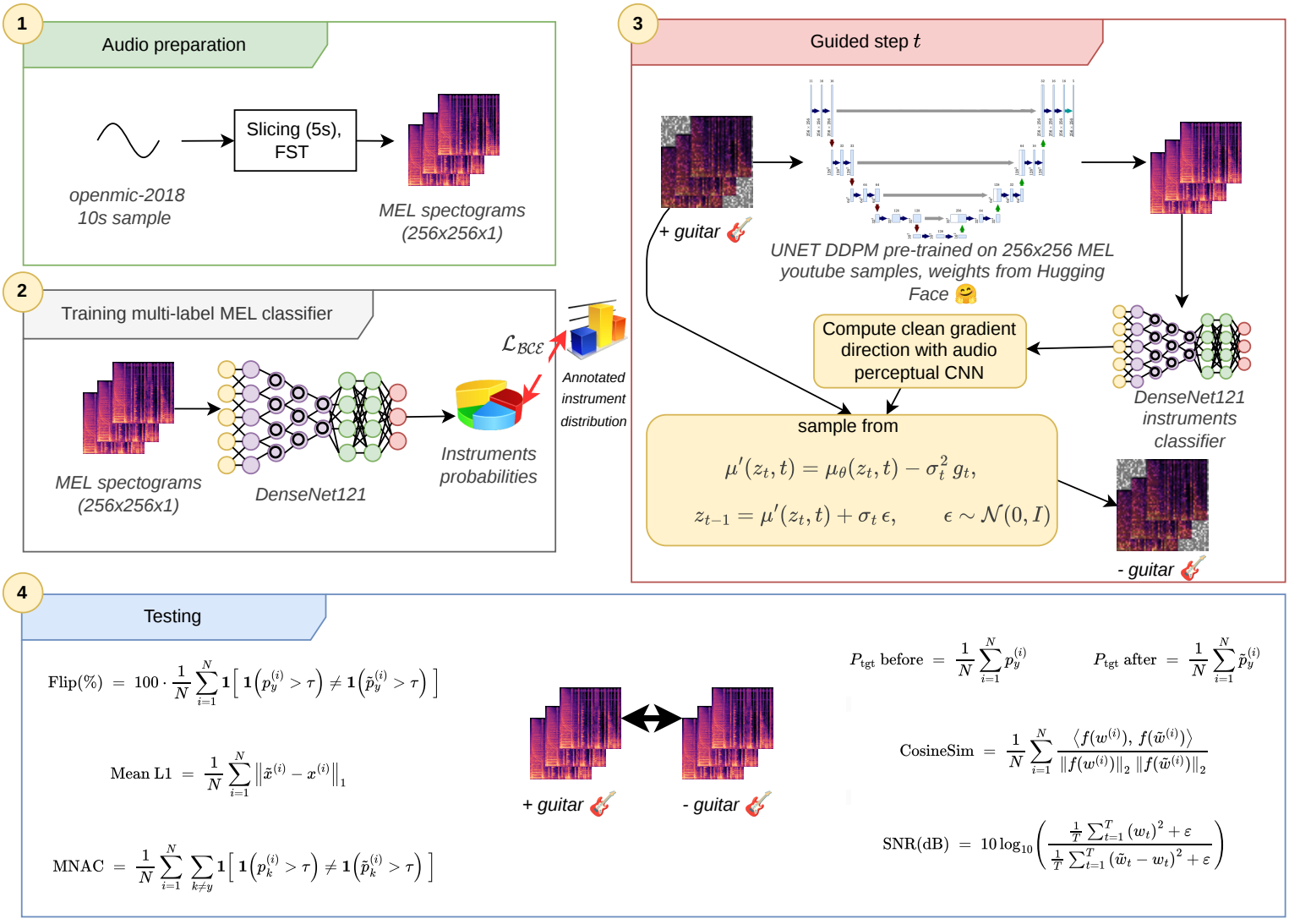
1
Audio preparation
Slicing (5s), FST
openmic-2018
10s sample
MEL spectograms
(256x256x1)
2
Training multi-label MEL classifier
$\mathcal{L}_{BCE}$
Annotated instrument distribution
MEL spectograms
(256x256x1)
DenseNet121
Instruments probabilities
3
Guided step t
+ guitar
UNET DDPM pre-trained on 256x256 MEL youtube samples, weights from Hugging Face
Compute clean gradient direction with audio perceptual CNN
DenseNet121 instruments classifier
sample from
$\mu'(z_t, t) = \mu_\theta(z_t, t) - \sigma_t^2\, g_t,$
$z_{t-1} = \mu'(z_t, t) + \sigma_t\, \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I)$
- guitar
4
Testing
$\mathrm{Flip}(\%) = 100 \cdot \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\left[\, \mathbf{1}\left(p_y^{(i)} > \tau\right) \neq \mathbf{1}\left(\tilde{p}_y^{(i)} > \tau\right) \,\right]$
$\text{Mean L1} = \frac{1}{N}\sum_{i=1}^{N} \left\| \tilde{x}^{(i)} - x^{(i)} \right\|_1$
$\mathrm{MNAC} = \frac{1}{N}\sum_{i=1}^{N} \sum_{k \neq y} \mathbf{1}\left[\, \mathbf{1}\left(p_k^{(i)} > \tau\right) \neq \mathbf{1}\left(\tilde{p}_k^{(i)} > \tau\right) \,\right]$
+ guitar
- guitar
$P_{\text{tgt}}\ \text{before} = \frac{1}{N}\sum_{i=1}^{N} p_y^{(i)}$
$P_{\text{tgt}}\ \text{after} = \frac{1}{N}\sum_{i=1}^{N} \tilde{p}_y^{(i)}$
$\text{CosineSim} = \frac{1}{N}\sum_{i=1}^{N} \frac{\langle f(w^{(i)}),\, f(\tilde{w}^{(i)})\rangle}{\|f(w^{(i)})\|_2\, \|f(\tilde{w}^{(i)})\|_2}$
$\text{SNR(dB)} = 10 \log_{10}\left(\frac{\frac{1}{T}\sum_{t=1}^{T} (w_t)^2 + \varepsilon}{\frac{1}{T}\sum_{t=1}^{T} (\tilde{w}_t - w_t)^2 + \varepsilon} \right)$